

ContentDepot[®] Edge[™]

Mike Pilone

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About NPR

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National Public Radio

- NPR is an independent, nonprofit media organization that was founded on a mission to create a more informed public.
- Approximately 99% of the U.S. population is within the broadcast listening area of one or more public radio stations.
- The network includes thousands of journalists at nearly 250 NPR Member stations across the country, in addition to the hundreds of journalists working at NPR.
- Today, NPR distributes content in many formats: articles, video, podcasts, and broadcast radio

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And Tiny Desk® concerts



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Distribution

NPR Distribution

- NPR Distribution manages the Public Radio Satellite System (PRSS), and provides satellite transmission services to regional public radio networks and commercial broadcasters across the United States.
- Primarily focuses on the broadcast radio portion of distribution
- Moves public radio content from producers to stations
- Distributes live audio streams as well as pre-recorded (file) programming

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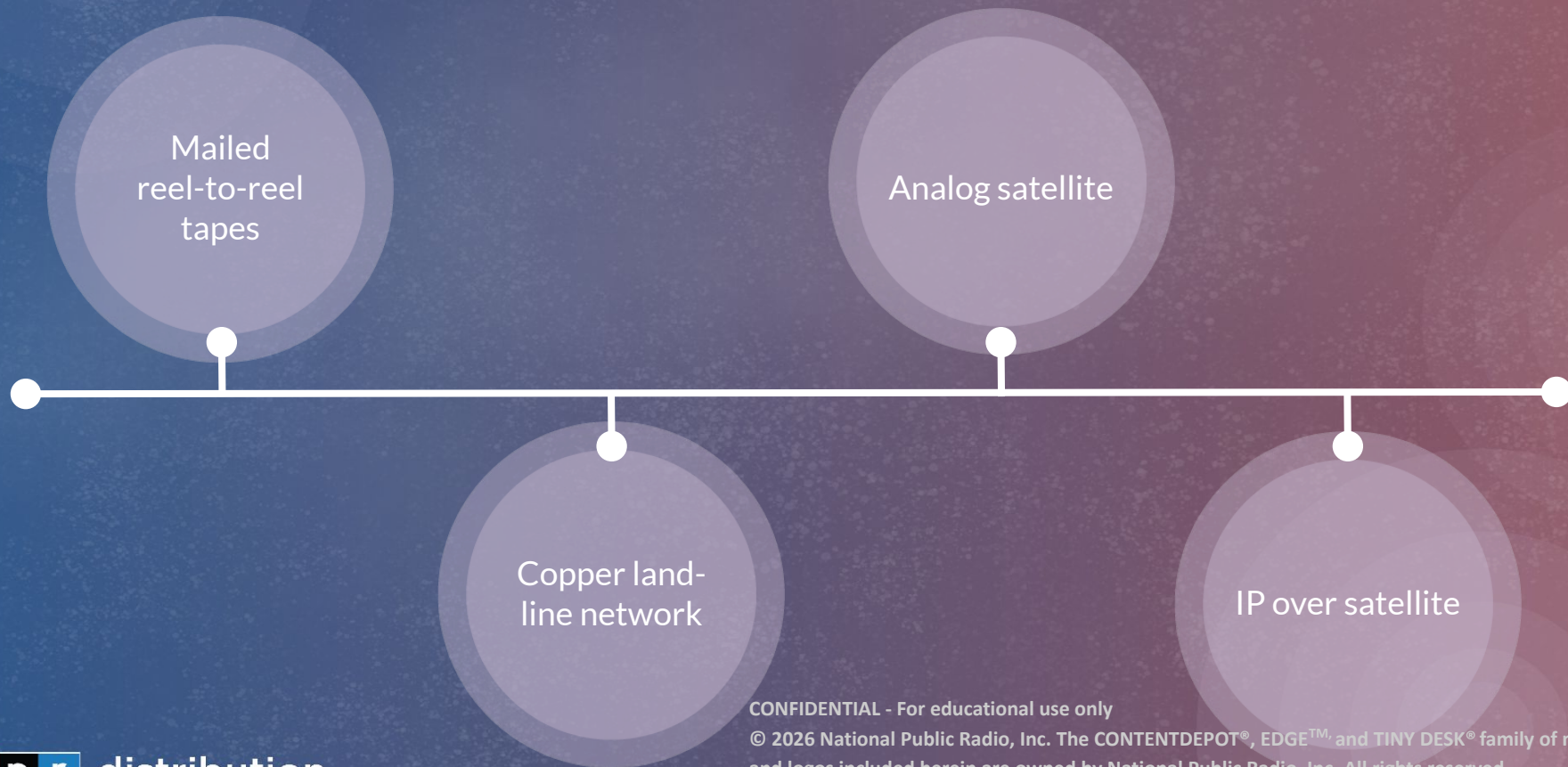
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Distribution Background

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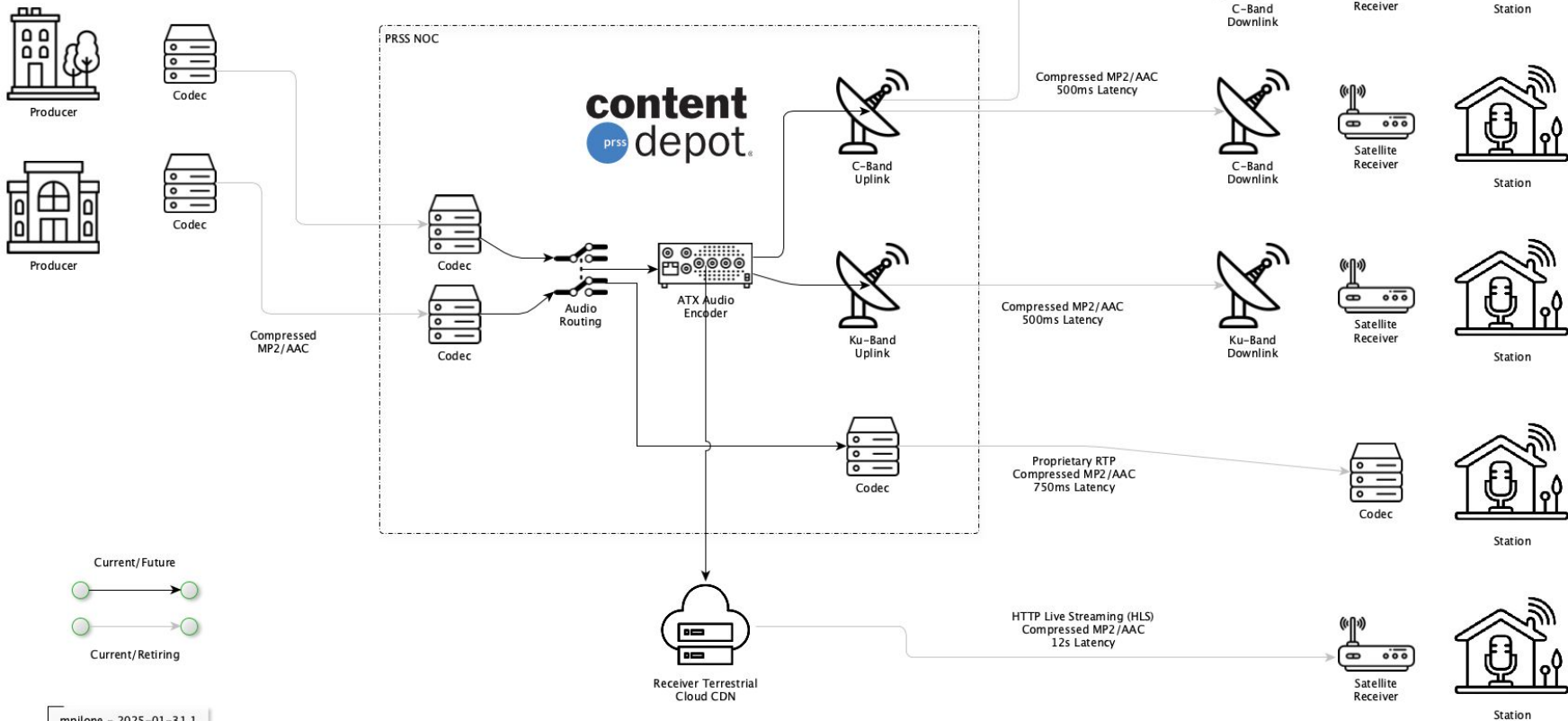
Distribution Technologies Over the Years



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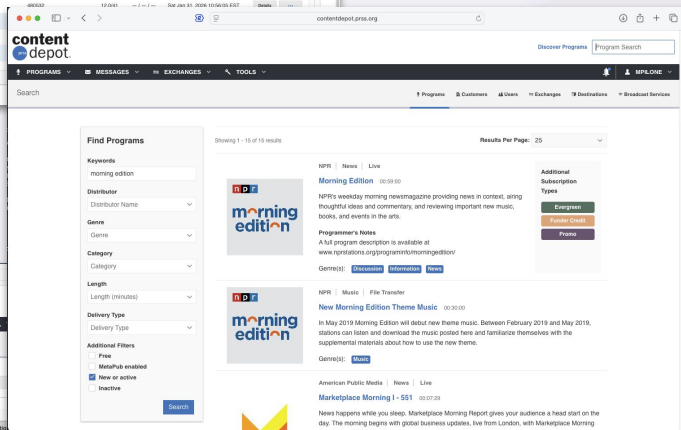
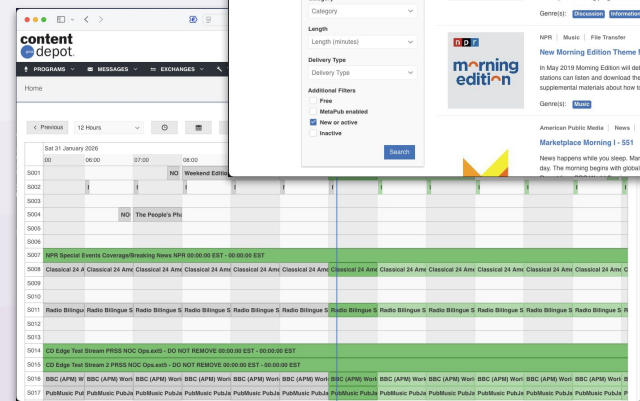
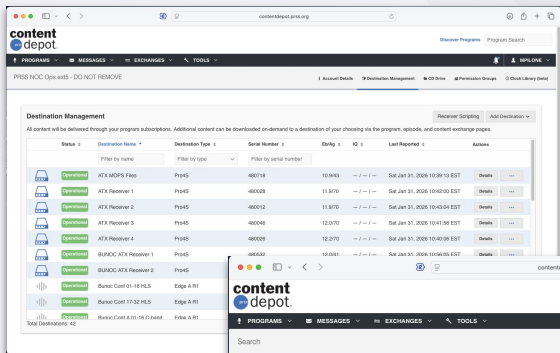
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How it Works Today



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ContentDepot®

- Scheduling system for stations
- Content distribution system for producers
- Our in-house technology
- Common interface for multiple distribution transports

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Satellite Integrated Receiver and Decoder

- Field programmable gate array (FPGA) and digital signal processor (DSP)
- 4 audio ports
- 2 rack units (RU) height
- 15 second HTTP live streaming (HLS)

fallback



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Challenges

- Satellite is primarily a 1-way technology
- Satellite is a one-to-many broadcast technology with limited bandwidth
- Satellite IRD hardware:
 - Out-of-date
 - Limited expandability
 - Expensive
- Shortage of RF engineers
- C-band satellite spectrum being reduced

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ContentDepot[®] Edge[™] Solution

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Analysis

Understanding the problems and goals

- Build vs buy
- Pain points of the current system
- Desired capabilities
- Technical evaluation
- Resource evaluation

Software Proof of Concept



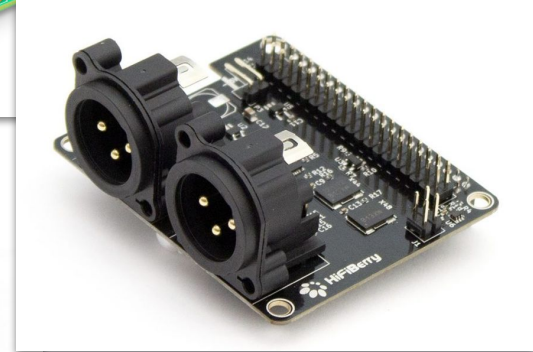
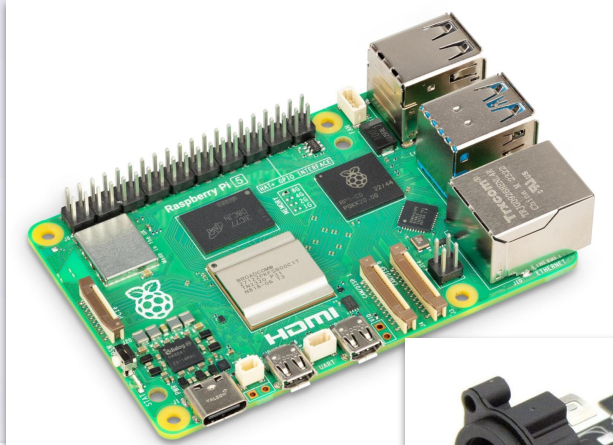
- Low cost, quick turnaround
- Encode and decode audio
- Network transport reliability
- Use open source solutions

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Audio Hardware Proof of Concept

- Use off-the-shelf components
- Raspberry Pi 5
- Audio hat
- Evaluate performance and scaling

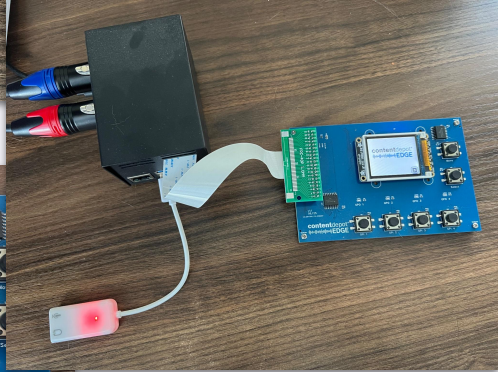
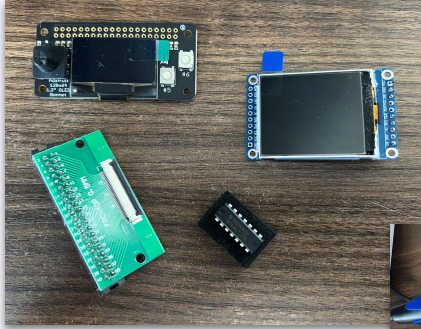
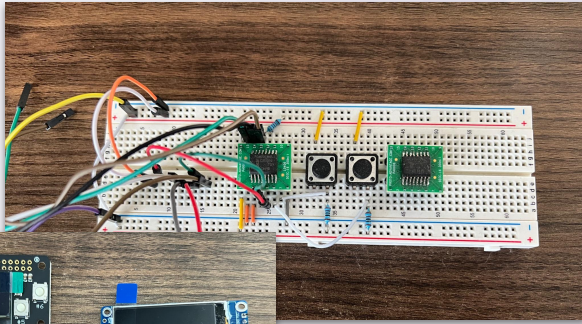


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Control Hardware Proof of Concept

- Breadboard basic functionality
- Control buttons
- Front panel display
- General purpose IO via LEDs
- Custom PCBs
 - PCBWay
 - JLCPCB



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Fleet Management

- Operation system build and maintenance
- Remote updates/firmware
- Monitoring
- Troubleshooting
- Inventory tracking



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Go / No-Go

ContentDepot[®] Edge[™] v1

- Custom audio board
- Decisions
 - Compute
 - Power
 - Storage
 - Feature set
- Enclosure
- Mockups and prototypes

Deploy and Evaluate

- Assembly
- Certification
- 18 station pilot
- Collect feedback, metrics

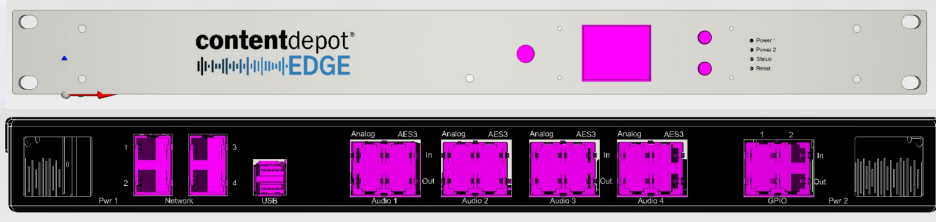
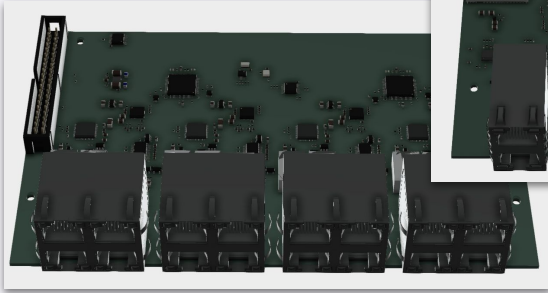
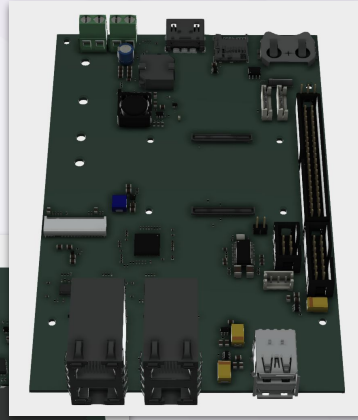


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Iterate and Improve

- Audio ports
- Network ports
- Compute module 5 (CM5)
- Power supplies
- Storage



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Lessons Learned

- Don't let perfection get in the way of progress
- Use proof of concepts and prototypes as much as possible
- Make use of rapid prototyping services
 - PCBs
 - Enclosures
 - CNC
 - Laser cutting
 - 3D printing
- There's probably a company out there that does exactly what you need
- Don't be afraid!

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Thank you!

Questions? Comments? Concerns?

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